Determining the Planck constant via the photoelectric effect and the Boltzmann constant via a to a Boltzmann factor proportional current at a p-n junction

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We measured the Planck constant to be $h = (6,26 \pm 0,57) \cdot 10^{-34}$ Js by using Einstein's equation for the photoelectric effect. We also measured the Boltzmann constant with the collector current in a transistor, which follows a voltage applied to base and emitter trough a Boltzmann factor. We get $k_B = (1.3745 \pm 0.0038) \cdot 10^{-23}\ \frac{\mathrm{J}}{\mathrm{K}}$.

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1 Introduction

The Planck constant h and Boltzmann constant k_B are one of the fundamental constants in modern physics. Planck introduced in 1900 the idea, that the energy of light is quantized, in order to explain the black-body radiation. Therefore he is considered to be the father of quantum mechanics. The energy of a light quantum according to Planck is the product of the frequency of the light and the Planck constant. So it connects the energy scale of light with the frequency scale. As in the discovery the Planck constant has a huge role in today's quantum mechanics. On the other hand Boltzmann constant connects the energy scale with the temperature scale. The Boltzmann constant plays an important role in statistical physics, where a description of each particle is not possible just because of the sheer number of particles. So a statistical statement about the system is made with respect to the energy of the particles and the temperature of the system. Because both constants are so important in today's physics, we wanted to determine the constants in this paper and check if our values are compatible with the literature values.

2 Measuring the Planck constant

2.1 Photoelectric effect

The photoelectric effect describes the phenomenon, that electrons are emitted when electromagnetic radiation, such as light, hits a material. After Planck the energy E of a photon with frequency ν is

$$E_{\mathrm{Ph}} = h\nu$$

So when light hits a material the energy of the photon is transferred to the electron. Therefore when the energy of the photon is smaller than the work function W_A of the material, the electron has not enough energy to leave the atom. This explains why the photoelectric effect only occurs with frequencies bigger than a critical frequency. When $E_{\mathrm{Ph}} > W_A$ the electron can be emitted and the remaining energy is in the kinetic energy of the electron. So one has

$$E_{kin} = h\nu - W_A \tag{1}$$

This equation goes back to Einstein in 1905. When the emitted electrons hit an anode, which is connected to the emitting material, the circuit is closed and a current can be measured. This is how a photo cell works. If one applies an opposing voltage, so that no current flows, the kinetic energy of the emitted electrons is equal to the product of the charge e of the electron and the applied voltage. With this method we measure the kinetic energy for different frequencies and can determine the Planck constant with (1).

2.2 Experiment and analysis

To determine the Planck constant we used a mercury lamp with different wavelengths filters in front of it to

illuminate a photodiode enclosed in a vacuum. To get a better signal we placed a light focusing lens in front of the light source. The counter voltage was applied to the photodiode in such a way that raising it will counteract the current from the photoeffect. The voltage was measured with a "Agilent 34405A" while the current was measured with a "Keithley 6485 Picoammeter". Even though the ammeter is build to have high precision the environment it is used in does not allow a precise measurement of the current. The error is therefore determined by the amount of fluctuations while measuring the current. A schematic of the experimental setup is shown in Fig. 1.

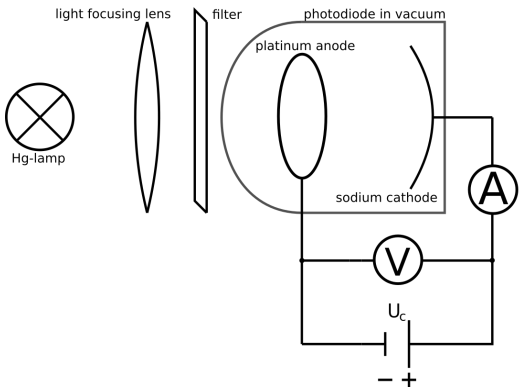

Fig. 1: Schematic setup to measure the Planck constant via the photoelectric effect. The light of the Hg-lamp goes trough a filter, so that almost only one wavelength hits the anode and causes the photoelectric effect, with which the Planck constant will be determined. The lens focuses the light, so that our measured current intensity is stronger. An opposing voltage was applied until no current flows, so that the kinetic energy of the emitted electrons can be determined.

Five different wavelength filters were used to determine the stopping voltage at which no current should flow. In theory the platinum anode should have a work function of $W_A = 5.32$ to 5.66 eV [2], which is higher than the energy of the photons used. Because of impurities in the anode, a photoeffect in the opposite direction can be seen after passing through the point at which no current should be flowing. Scattered light can still hit the anode and is able to ionize electrons. So taking the point at which no current is flowing as the stopping voltage is no longer sensible. One way to still determine the stopping voltage for the sodium cathode is to assume that the measured photoeffect for the impurities in the anode is for the most part linear in the region we expect the stopping voltage of sodium to be. We can then determine the point at which the linear part of the data switches into the non-linear part. Assuming we have enough light intensity shining on the photodiode so that we are in the region, where Child's law can be applied, we can expect the current to be proportional to the voltage as followed:

$$I(U) \propto U^{\frac{3}{2}} \tag{2}$$

By fitting a piecewise-defined function to our data, we have a systematic way to determine the described switching point. The function is a linear function up to a critical value of U_0, which determines the stopping value, and from then on increases with $(U - U_C)^{\frac{3}{2}}$. We also tried to determine the switching point by eye to have a way to evaluate the selected stopping voltages for the first method. For the fit method we also used the errors that were given. The error for the values determined by us are depending on the uncertainty of the interval in which we would put the cutoff point in which the linear part switches to the non-linear part. One example for the wavelength $\lambda = 366$ nm is shown in Fig. 2. The figures for the other graphs are attached in the appendix.

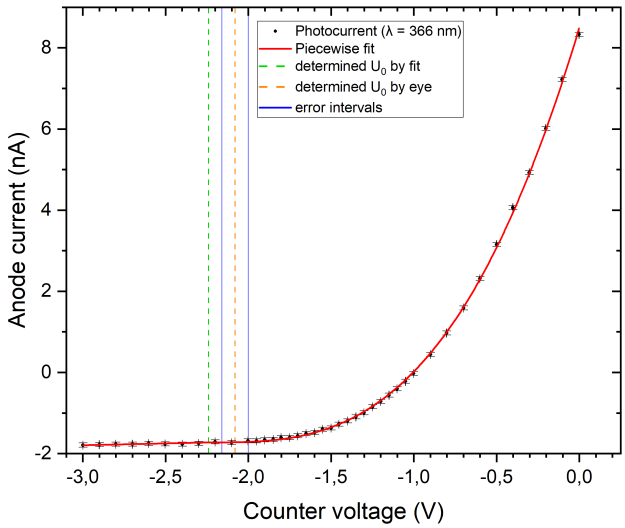

Fig. 2: The measured current plotted against the applied counter voltage for a wavelength of 366 nm. The current is unequal to zero for high counter voltages, because a photoelectric effect at the anode occurs. This makes finding the right voltage difficult, so one must guess, where the region for Child's law start. Another method is using the fit of a piecewise-defined function to the data. The value determined by the piecewise fit is $U_0 = (2,245 \pm 0,011)$ V. The value determined by eye is $U_0 = (2,08 \pm 0,10)$ V. The error interval shown is from the the value determined by eye. Only for this wavelength the values differ by more than the margin of error.

In table 1 the values for the determined stopping voltages for both methods are listed.

The determined value coincide within the error of margin with one another except the value for a wavelength of $\lambda = 366$ nm. We decided to use both methods to determine the Planck constant and see if they are consistent with each other.

wavelength	U_0/V by fit	U_0/V by eye
366 nm	2.245 ± 0.011	2.08 ± 0.10
405 nm	1.6258 ± 0.0095	1.600 ± 0.040
436 nm	$1,.299 \pm 0.0071$	1.410 ± 0.060
546 nm	0.9333 ± 0.0046	0.920 ± 0.080
578 nm	0.8108 ± 0.0048	0.79 ± 0.10

Table 1: The determined counter voltage, which suppresses the electrons, which are emitted by the photoelectric effect at the cathode, to reach the anode, for different wavelengths causing the photoelectric effect. One value was determined by looking at the graphs, the other by fitting a piecewise-defined function to the data.

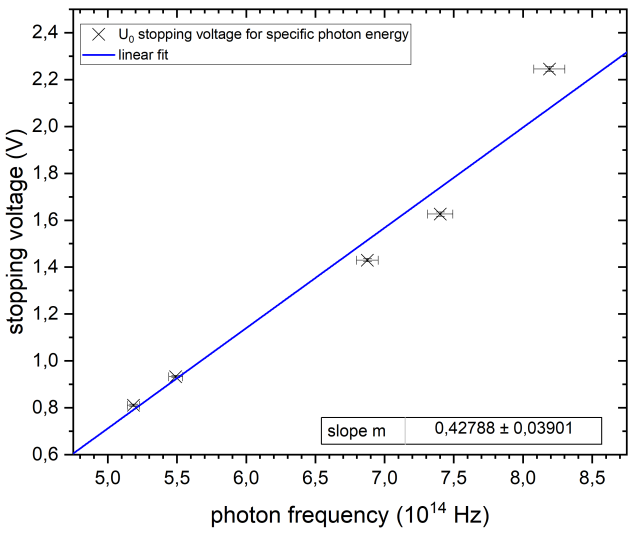

Fig. 3: The stopping voltages, which where determined by fitting a piecewise-defined function to the data, plotted against the frequency of the photon causing the photoelectric effect. A linear function was fitted, so one can determine the Planck constant trough the slope using Einstein's equation for the photoeffect.

The stopping voltages were plotted in dependency of the frequency ν of the photons. A linear fit was performed (see Fig. 3 and Fig. 12 in the appendix). From formula (1) one can determine h by multiplying the elementary charge e with the slope m of the linear fit. Doing so we get the following values:

$$h_{fit} = (6.86 \pm 0.63) \cdot \ 10^{-34} \ \text{Js}$$

$$h_{eye} = (6.26 \pm 0.57) \cdot \ 10^{-34} \ \text{Js}$$

Both values coincide within the margin of error with one another and with the literature value for the exactly defined Planck constant $h = 6.62607015 \cdot \ 10^{-34}$ Js [3]. The large error is due to the low number of data points collected. Measurements for more wavelengths would allow a more precise determination of the Planck constant with smaller uncertainty. Because of systematic errors from impurities in the anode and factors such as the contact voltages between the used materials, the determination of the stopping voltages was the biggest challenge. Considering all difficulties the methods used seem to give plausible results.

3 Measuring the Boltzmann constant

3.1 Theory behind the method used to determine the Boltzmann constant

When a n doped and a p doped semiconductor are combined, electrons and holes recombine due to diffusion and a depletion zone with no carrier of charge develops. In this region an electric filed from the atoms, which provided the electrons and holes, is created, which prevents a recombination of all electrons and holes. This means, that a potential difference U_A arises in opposite direction to the diffusion. In the depletion zone electrons are thermally excited and then accelerated because of the electric field. This forms a intrinsic current I_0. Electrons with an energy $E > eU_\text{A}$ can overcome the depletion zone and recombine with holes. That means a current I_D flowing opposing to I_0. Because the electrons in the conduction band follows the Boltzmann statistic, one has

$$I_\text{D}(U_\text{A}) \propto e^{\frac{eU_\text{A}}{k_B T}}$$

In thermal equilibrium with no outer voltage there should not be any current, so it follows

$$|I_\text{D}(U_\text{A})| = |I_0|$$

When an outer voltage U is applied, the anti diffusion voltage decreases. So the diffusion current is

$$I_\text{D}(U) = |I_0| e^{\frac{eU}{k_B T}} \tag{3}$$

So when everything listed is considered, one gets the Shockley diode equation

$$I(U) = |I_0|(e^{\frac{eU}{k_B T}} - 1) \tag{4}$$

With (3) the Boltzmann constant k_B can be determined by measuring the current flowing trough a diode, when an outer voltage is applied, so that the p-n junction is forward biased. [1]

But in reality there are currents flowing on the boundaries of the crystal, which distort the measurement. So we will use a setup, shown in Fig. 4, with a npn transistor, which was heated by the thermostat 'Ecoline Startedition RE104' from Lauda. So the pure diffusion current is measured at the collector, whereas I_0 of each n-p junction compensates each other and the disturbing current flows to the basis, because no voltage is applied between basis and collector.

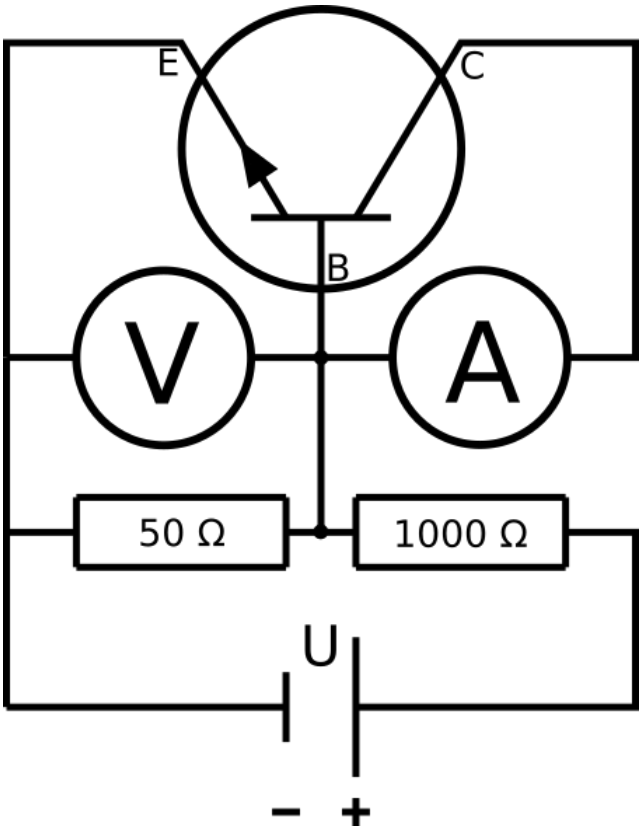

Fig. 4: Circuit for measuring the Boltzmann constant. We used a npn transistor and applied a voltage to base and emitter. The collector current follows the voltage trough a Boltzmann factor, so we can calculate the Boltzmann constant. A divider circuit was added, so the voltage could be applied more precisely.

3.2 Experiment and analysis

We measured the collector current I_C in dependency of the applied basis emitter voltage U_BE for a transistor temperature of 40 °C, 80 °C and 120 °C. For both measurements the "Agilent 34405A" was used. The observed data for $T = 40°\mathrm{C}$ with a fit of function (3) and its residues is shown in Fig. 5. Because the residuals look the same for the three temperatures, we suspect to have a systematic error in our measurements. The base emitter resistance is not negligible, so U in (3) is smaller than our measured base emitter voltage.

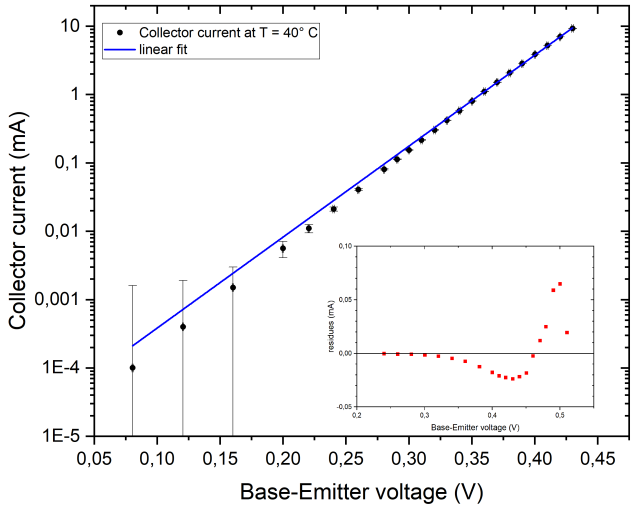

Fig. 5: Collector current of a transistor at $T = 40°\mathrm{C}$ in dependency of the applied voltage between base and emitter with an exponential fit to the data. The residuals show a systematic error in our measurement.

To get a grip of the systematic error we used a R which is the resistance that accounts for the voltage that is overrated. The formula that follows is:

$$I_\mathrm{D}(U) = |I_0| e^{\frac{e(U - R \cdot I_C)}{k_B T}} \tag{5}$$

Rearranging the equation gives us a function in which R appears as a parameter which will be fitted.

$$U = \frac{k_B \cdot T}{e} \cdot \ln(\frac{I}{I_0}) + R \cdot I_C \tag{6}$$

From this rough estimate of the resistance we can correct the measured voltage and reduce the systematic error. For the measurement at $T = 40°C$ the fitted function and the value for R are shown in Fig. 6.

After correcting the voltage we first tried to fit all data points, but a systematic error is still visible. Although it is at least one magnitude smaller we still decided to exclude data points in which the systematic error is still significant, because those data points have a smaller relative error so they are being weighted more heavily by the fit which is not desired. So by taking only the data points in which the systematic error is smaller we expect a more accurate value. The residues when fitting all data points for all three temperatures with the corrected voltages are attached in the appendix (see Fig. 13, 14 and 15). Fig. 7 shows the fit and residues for $T = 40°\mathrm{C}$. The figures for the other two temperatures are also in the appendix (Fig. 16 and 17).

The values for the Boltzmann constant for the different temperatures of the transistor are shown in table 2.

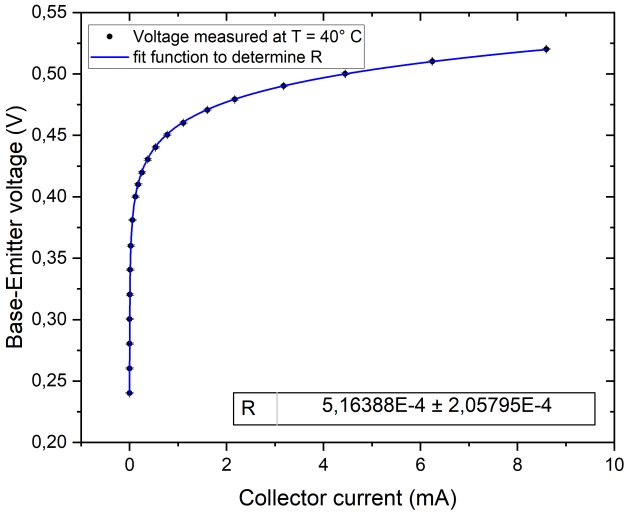

Fig. 6: The applied voltage between base and emitter in dependency of the collector current of a npn transistor. A function was fitted to determine the resistance trough a fit parameter.

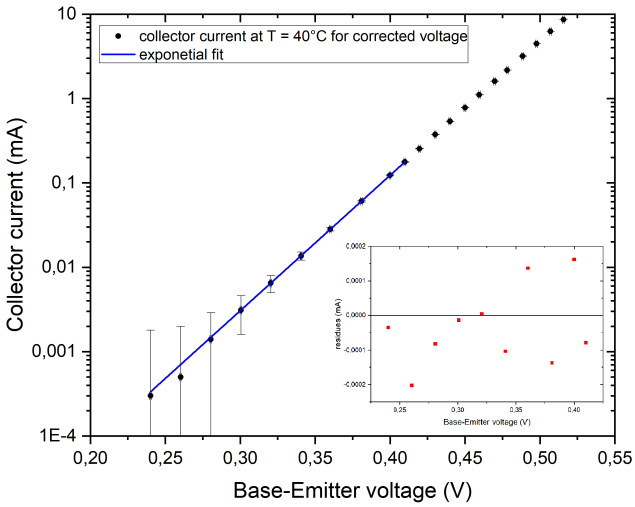

Fig. 7: Final fit after determining which data points to fit because of the systematic error seen in the residues of fitting all data points (see Fig. 13). The slope m = $(36,959 \pm 0,048)\ \frac{\mathrm{C}}{\mathrm{J}}$ gives us the value to determine the Boltzmann constant. The residues are smaller than the residues of the uncorrected fits, indicating a reduction of the systematic error from before.

transistor temperature $T/°\mathrm{C}$	$k_B/\frac{\mathrm{J}}{\mathrm{K}} \cdot 10^{-23}$
40.00 ± 0.40	1.384 ± 0.014
80.00 ± 0.40	1.3763 ± 0.0073
120.00 ± 0.40	1.3727 ± 0.0047

Table 2: The calculated Boltzmann constants for different temperature of the transistor

Only the value for the temperature $T = 40°\mathrm{C}$ is not coinciding with the exactly defined value of the Boltzmann constant $k_B = 1.380649 \cdot 10^{-23}\frac{\mathrm{J}}{\mathrm{K}}$ [3]. The weighted mean with error is also calculated:

$$k_B = (1.3745 \pm 0.0038) \cdot 10^{-23}\,\frac{\mathrm{J}}{\mathrm{K}}$$

This value is not coinciding within its margin of error with the exact value for the Boltzmann constant.

The systematic errors due to the behavior of the transistor especially at higher voltages can not be fully eliminated. It is possible that at higher voltages we reached the saturation region of the transistor. The estimated value for the resistance R is not constant because we are not dealing with a ohmic resistor. Rather a differential resistance for the Base-Emitter voltage drop is expected.

4 Conclusion

We measured the Planck constant h via the photoelectric effect. Therefore we determined the stopping voltage for the electron once by eye and once by fitting a piecewise-defined function to the data. The calculated values for the Planck constant are $h_{fit} = (6.86 \pm 0.63) \cdot 10^{-34}$ Js, $h_{eye} = (6.26 \pm 0.57) \cdot\ 10^{-34}$ Js. Both values are compatible with the literature value. For determining the Boltzmann constant we measured the collector current of a transistor in dependency of the voltage applied between base and emitter for different temperatures. For $T = 40°\mathrm{C}$ we got $(1.384 \pm 0.014) \cdot 10^{-23}\ \frac{\mathrm{J}}{\mathrm{K}}$, for $T = 80°\mathrm{C}$ we got $(1.3763 \pm 0.0073) \cdot 10^{-23}\ \frac{\mathrm{J}}{\mathrm{K}}$, for $T = 120°\mathrm{C}$ we got $(1.3727 \pm 0.0047) \cdot 10^{-23}\ \frac{\mathrm{J}}{\mathrm{K}}$. Because of systematic errors in our measurement, only two of the there values coincide within the margin of error with the exact value for the Boltzmann constant. The mean value does also not coincide within the margin of error so another setup needs to be used for measuring the Boltzmann constant more precisely.

References

[1] Physikalisches Grundpraktikum, Universität Würzburg, Modul C2, Versuch 45, Plancksches Wirkungsquantum und Boltzmann-Konstante, 2022

[2] Work function of Platinum, FH Münster, visited on 04/06/2022, https://www.fh-muenster.de/ciw/downloads/personal/juestel/juestel/chemie/Austrittsarbeit.pdf

[3] Value for the Planck constant and the Boltzmann constant, E. Tiesinga et al., "CODATA recommended values of the fundamental physical constants: 2018", Rev. Mod. Phys. 93, 025010 (2021)

5 Appendix

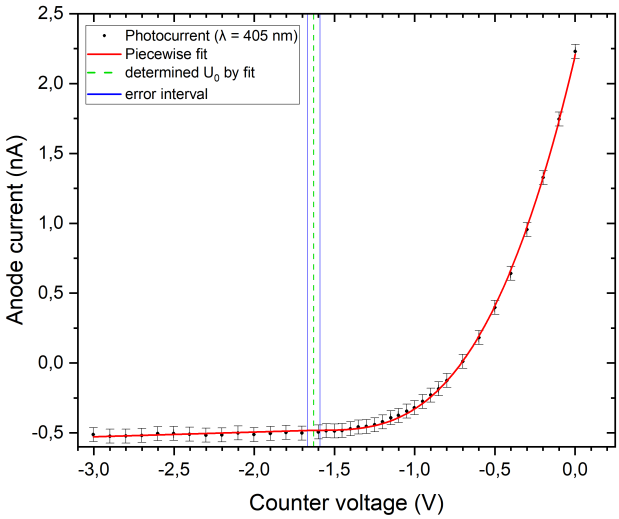

Fig. 8: The measured current plotted against the applied counter voltage for a wavelength of 405 nm for finding the stopping voltage, so one can calculate the Planck constant. The value determined by the piecewise fit and by eye are so close that only the fit value is shown. It is $U_0 = (1,6258 \pm 0,0095)$ V. The error interval shown is from the the value determined by eye.

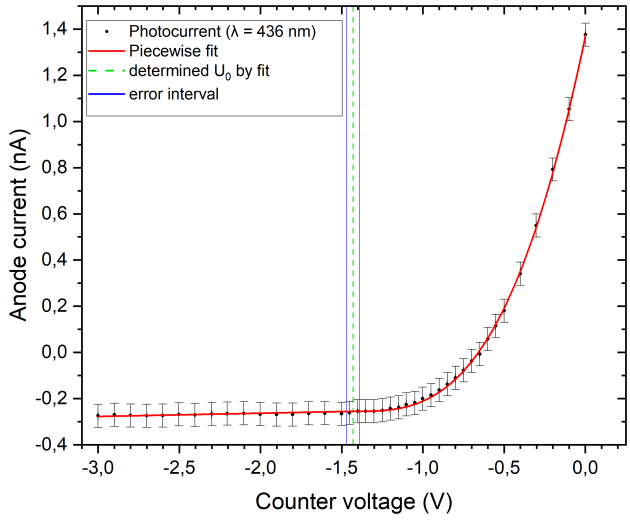

Fig. 9: The measured current plotted against the applied counter voltage for a wavelength of 436 nm for finding the stopping voltage, so one can calculate the Planck constant. The value determined by the piecewise fit and by eye are so close that only the fit value is shown. It is $U_0 = (1,4299 \pm 0,0071)$ V. The error interval shown is from the the value determined by eye.

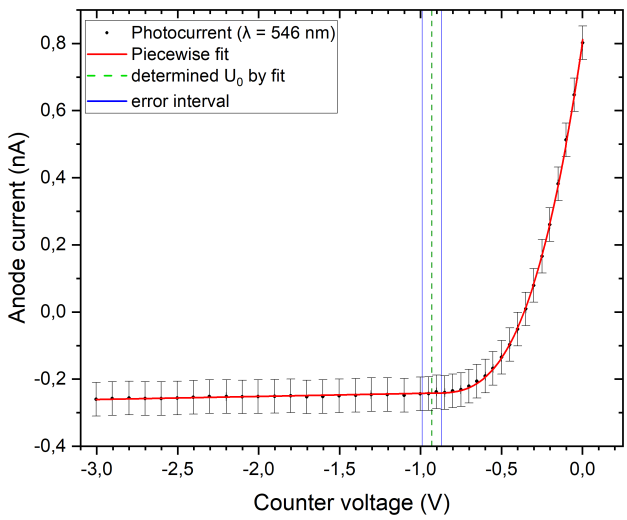

Fig. 10: The measured current plotted against the applied counter voltage for a wavelength of 546 nm for finding the stopping voltage, so one can calculate the Planck constant. The value determined by the piecewise fit and by eye are so close that only the fit value is shown. It is $U_0 = (0,9333 \pm 0,0046)$ V. The error interval shown is from the the value determined by eye.

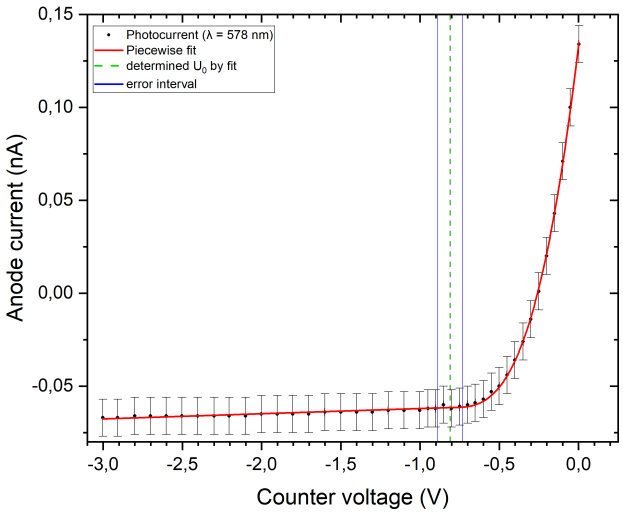

Fig. 11: The measured current plotted against the applied counter voltage for a wavelength of 578 nm for finding the stopping voltage, so one can calculate the Planck constant. The value determined by the piecewise fit and by eye are so close that only the fit value is shown. It is $U_0 = (0,8108 \pm 0,0048)$ V. The error interval shown is from the the value determined by eye.

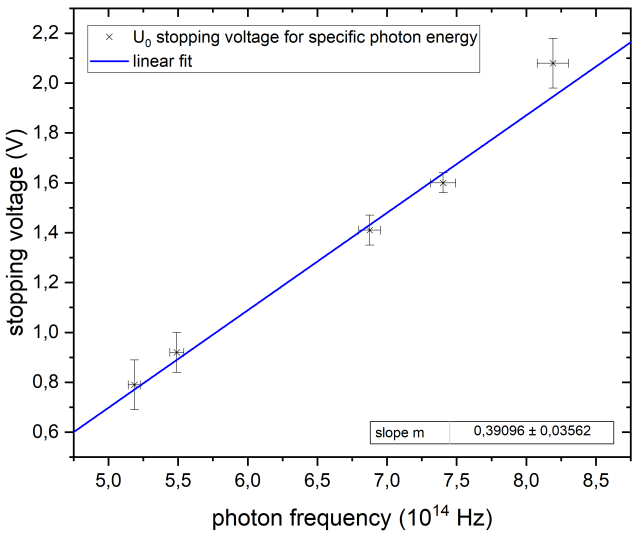

Fig. 12: The stopping voltages, which where determined by eye, plotted against the frequency of the photon causing the photoelectric effect. A linear function was fitted, so one can determine the Planck constant trough the slope after Einstein's equation.

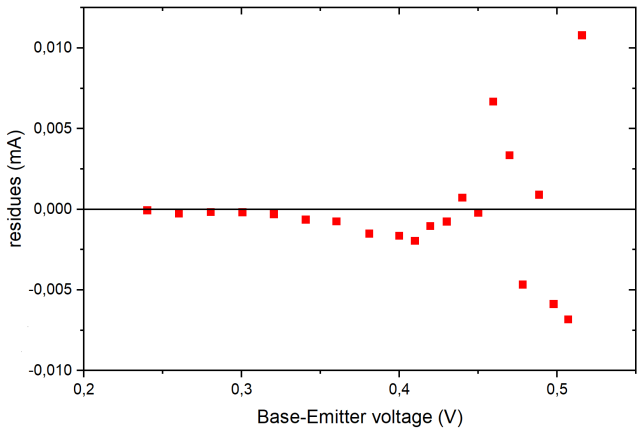

Fig. 13: Residues for T = 40 °C when fitting all data points after correcting the voltages. A systematic error is still noticeable, but compared to before the residues seem to have gotten smaller indicating a better fit and reduction of the systematic error.

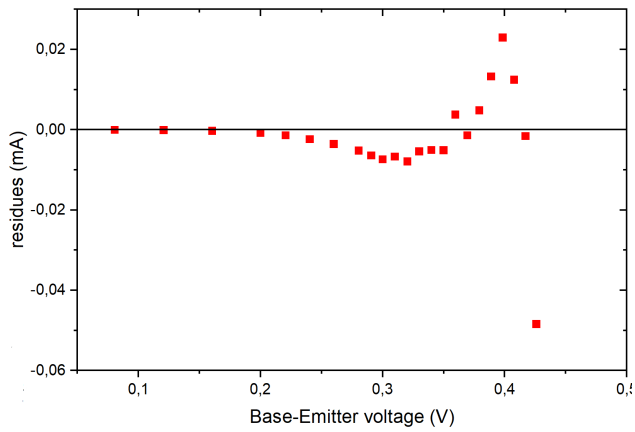

Fig. 14: Residues for T = 80 °C when fitting all data points after correcting the voltages. A systematic error is still noticeable, but compared to before the residues seem to have gotten smaller indicating a better fit and reduction of the systematic error.

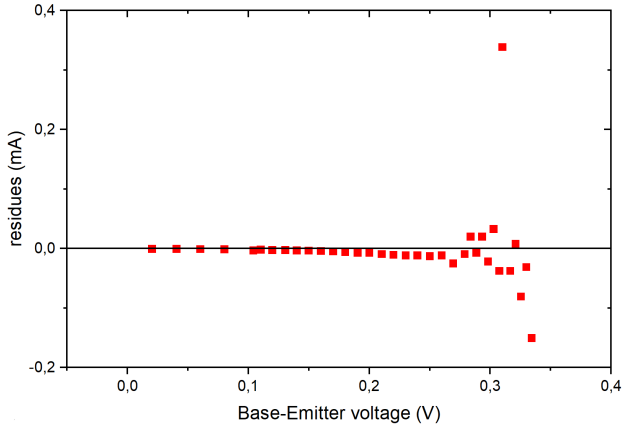

Fig. 15: Residues for T = 120 °C when fitting all data points after correcting the voltages. A systematic error is still noticeable, but compared to before the residues seem to have gotten smaller indicating a better fit and reduction of the systematic error.

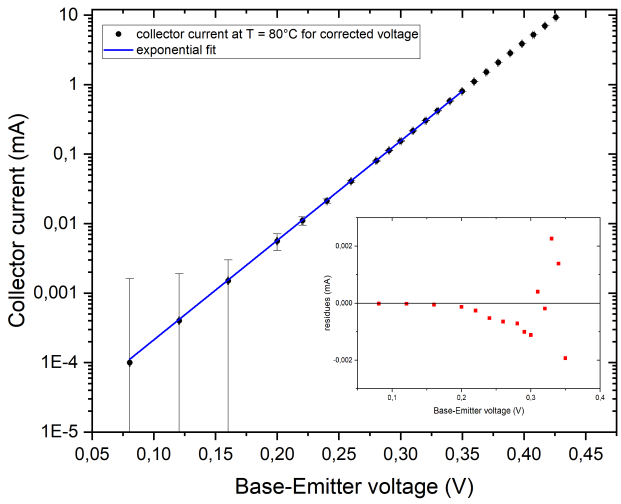

Fig. 16: Final fit after determining which data points to fit because of the systematic error seen in the residues of fitting all data points (see Fig. 14). The slope m = $(32,963 \pm 0,059)\ \frac{C}{J}$ gives us the value to determine the Boltzmann constant. The residues are smaller than the residues of the uncorrected fits, indicating a reduction of the systematic error from before.

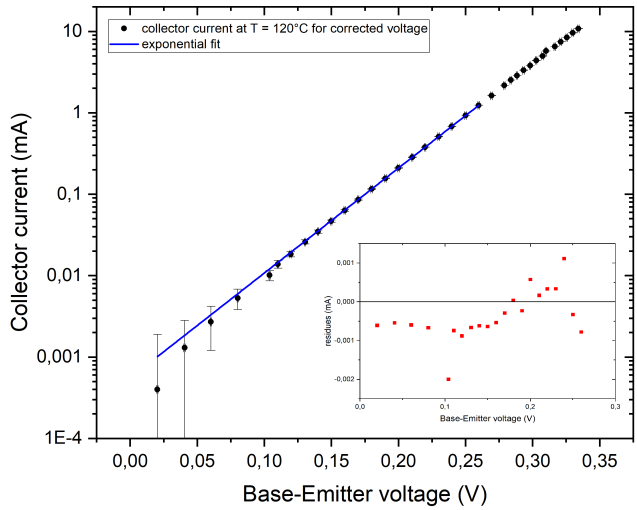

Fig. 17: Final fit after determining which data points to fit because of the systematic error seen in the residues of fitting all data points (see Fig. 15). The slope m = $(29,687 \pm 0,026)\ \frac{C}{J}$ gives us the value to determine the Boltzmann constant. The residues are smaller than the residues of the uncorrected fits, indicating a reduction of the systematic error from before.